Assignment
Data Mining & Data Warehousing
(DS7E-702)

1. What is the primary purpose of data warehouse?

The primary purpose of a data warehouse is to collect, integrate, and store large volumes of historical data from multiple heterogeneous sources in a centralized repository. It is designed to support decision-making processes by enabling complex queries, reporting, and analysis. Unlike operational systems, a data warehouse focuses on providing a consolidated view of data for strategic planning, trend analysis, and business intelligence.

2. How does the data warehouse different from a standard database?

A data warehouse differs from a standard database in terms of purpose, design, and usage. A standard database (OLTP) is optimized for handling real-time transactions such as insertions, updates, and deletions, ensuring data consistency and speed. In contrast, a data warehouse (OLAP) is optimized for querying and analyzing large datasets. It stores historical data, is subject-oriented, time-variant, and non-volatile, whereas standard databases focus on current operational data.

3. Explain the ETL process.

The ETL process stands for Extract, Transform, and Load. It is a critical component in building a data warehouse.

- **Extract:** Data is collected from various sources such as databases, files, or APIs.
- **Transform:** The extracted data is cleaned, filtered, aggregated, and converted into a consistent format.
- **Load:** The transformed data is loaded into the data warehouse for analysis.

This process ensures data quality, consistency, and usability.

4. What is the difference between structured and unstructured data?

Structured data is highly organized and stored in a predefined format such as tables with rows and columns (e.g., relational databases). It is easy to query and analyze using SQL.

Unstructured data, on the other hand, does not follow a specific format and includes data such as text documents, images, videos, and social media content. It requires advanced processing techniques like natural language processing and machine learning for analysis.

5. What is a Data Mart? Give an example.

A Data Mart is a subset of a data warehouse that focuses on a specific business area or department, such as sales, marketing, or finance. It contains summarized or filtered data relevant to that domain, making it easier and faster to analyze.

Example: A sales data mart may include data on customer purchases, revenue, and regional sales performance.

6. What are fact and dimension tables? And what is the difference b/w them?

Fact and dimension tables are key components of a data warehouse schema.

- **Fact Table:** Contains quantitative data such as sales amount, revenue, or transaction counts.
- **Dimension Table:** Contains descriptive attributes such as customer name, product category, or time.

The fact table is linked to dimension tables through foreign keys, enabling multidimensional analysis.

7. What is metadata? Give an example.

Metadata is defined as "data about data." It provides information about the structure, format, and meaning of data stored in a system.

Example: Table names, column definitions, data types, and relationships between tables in a database schema.

8. Difference between OLTP and OLAP?

OLTP (Online Transaction Processing) systems are used for managing real-time operations such as banking transactions and order processing. They are optimized for speed, concurrency, and data integrity.
OLAP (Online Analytical Processing) systems are used for analysing historical data and supporting decision-making. They allow complex queries, aggregations, and trend analysis.
While OLTP systems handle frequent updates, OLAP systems are read-heavy and designed for multidimensional analysis. OLTP uses normalized schemas, whereas OLAP often uses denormalized schemas like star or snowflake.

9. What is schema in a data warehouse and its types?

- A schema defines the logical structure of a data warehouse. It determines how data is organized and related.
- **Star Schema:** Central fact table connected to dimension tables; simple and efficient.

- **Snowflake Schema:** Normalized version of star schema; reduces redundancy but increases complexity.
- **Galaxy Schema:** Multiple fact tables sharing dimension tables.
  These schemas help optimize query performance and data organization.

These schemas help organize data efficiently for analysis.

10. What is a BI tool?

A Business Intelligence (BI) tool is software used to analyze data, generate reports, and visualize insights through dashboards and charts. It helps organizations make informed decisions by transforming raw data into meaningful information. Examples include Power BI, Tableau, and Qlik.

11. What is the simplest definition of data mining?

Data mining is the process of discovering hidden patterns, relationships, and useful information from large datasets using statistical, mathematical, and machine learning techniques.

12. Describe the iterative nature of KDD.

KDD (Knowledge Discovery in Databases) is an iterative process involving multiple steps: data selection, preprocessing, transformation, data mining, and evaluation. Each step may be repeated to improve results. For example, after evaluation, data may need further cleaning or transformation. This iterative nature ensures better accuracy and meaningful insights.

13. What is data smoothing? And which theoretical technique are used to handle "Noisy Data"?

Data smoothing is used to remove noise from datasets to improve data quality. Noisy data contains errors or random variations. Techniques include:

- **Binning:** Grouping values and smoothing them.
- **Regression:** Fitting a function to the data.
- **Clustering:** Detecting and removing outliers.

These methods help in producing more accurate models.

14. In association rule mining, what is the mathematical difference b/w support and confidence?

Support measures how frequently an itemset appears in the dataset. Confidence measures the probability that a rule is true when a condition is met.
For example, if people who buy bread also buy butter:

- Support shows how often both are bought together.
- Confidence shows how likely butter is bought when bread is purchased.

15. Explain the "centroid" concept in k-means clustering.

In k-means clustering, a centroid represents the center of a cluster. It is calculated as the mean of all points in the cluster. The algorithm assigns data points to the nearest centroid and updates centroids iteratively until convergence. This helps in grouping similar data points effectively.

16. What are the theoretical limitations of the Apriori algorithm regarding large data sets?

The Apriori algorithm is widely used for association rule mining, but it has several limitations, especially with large datasets. One major issue is its high computational complexity, as it generates many candidates item sets. It requires multiple scans of the database, which increases processing time and resource usage.

Another limitation is inefficiency when dealing with low support thresholds, as this leads to an explosion of candidate sets. Additionally, it does not scale well for big data environments.

To overcome these issues, improved algorithms like FP-Growth have been developed, which reduce database scans and improve performance.

17. What is "Overfitting" in predictive modelling, and how does "Pruning" address this in Decision Trees?

Overfitting occurs when a model learns not only the underlying patterns in the data but also noise and irrelevant details. As a result, it performs well on training data but poorly on unseen data.

In decision trees, overfitting leads to very complex trees with many branches.

Pruning is used to address this issue by removing unnecessary branches. There are two types:

- **Pre-pruning:** Stops tree growth early.
- **Post-pruning:** Removes branches after full tree construction.
  Pruning improves model generalization, reduces complexity, and enhances predictive accuracy.

18. What is Market Basket Analysis?

- Market Basket Analysis is a data mining technique used to identify relationships between items purchased together. It is widely used in retail to understand customer buying behaviour.

- For example, if customers frequently buy bread and butter together, businesses can place them near each other or offer combo deals.
- It uses association rule mining techniques like Apriori and FP-Growth. Metrics such as support, confidence, and lift are used to evaluate relationships. This technique helps in cross-selling, recommendation systems, and improving sales strategies.

19. Difference b/w Supervised vs Unsupervised learning?

Supervised learning uses labelled datasets where input-output pairs are known. It is used for classification and regression tasks.
Unsupervised learning works with unlabelled data and aims to find hidden patterns or clusters.

Examples:

- Supervised: Decision Trees, SVM
- Unsupervised: K-means, Hierarchical clustering

20. What is Bayesian classification?

- Bayesian classification is based on Bayes' Theorem, which calculates the probability of a class given certain feature. It assumes that features are conditionally independent.
- The most common form is the Naïve Bayes classifier, which is simple, fast, and effective for large datasets.
- It is widely used in spam filtering, text classification, and medical diagnosis.
- Despite its simplicity, it performs well even with limited data and handles uncertainty effectively.

21. Explain Markov Chain model.

- A Markov Chain is a stochastic model where the next state depends only on the current state, not on previous states (memoryless property).
- It is represented using states and transition probabilities.
- Applications include weather prediction, stock market analysis, and natural language processing.
- It simplifies complex systems by focusing only on present conditions.

22. Describe the "Hierarchical Clustering" method.

Hierarchical clustering is a method of clustering that builds a hierarchy of clusters. It can be:

- **Agglomerative (bottom-up):** Starts with individual points and merges them.
- **Divisive (top-down):** Starts with one cluster and splits it.
  It is represented using a dendrogram, which shows cluster relationships.
  Unlike k-means, it does not require the number of clusters in advance.
  However, it is computationally expensive for large datasets.

23. What is Data Preprocessing?

Data preprocessing involves preparing raw data for analysis. Steps include:

- Data cleaning (handling missing values)
- Data integration (combining sources)
- Data transformation (normalization)
- Data reduction (reducing size)
  It improves data quality and model accuracy.

24. Define the technique for data cleaning data, Data reduction?

Data cleaning involves detecting and correcting errors, handling missing values, and removing noise. Techniques include imputation, smoothing, and outlier detection.
Data reduction reduces data size while maintaining its integrity. Techniques include:

- Dimensionality reduction (PCA)
- Data compression
- Sampling

These techniques improve efficiency and reduce computational cost.

25. What is Data cube aggregation and dimensionality reduction?

Data cube aggregation involves summarizing data across multiple dimensions to provide higher-level insights. For example, sales data can be aggregated by region, time, or product category.
Dimensionality reduction reduces the number of variables in a dataset while preserving important information. Techniques like PCA and feature selection are commonly used.
These methods improve performance, reduce storage requirements, and simplify data analysis.